Machine Learning Interview Notes

Bias, Variance, Underfitting & Overfitting

Prepared from this chat for quick interview preparation and GitHub documentation.

1. Variance

Simple definition: Variance tells us how much a model's predictions change when the training data changes.

In simple words: Variance tells us how sensitive the model is to the particular training data it saw.

High variance: A small change in training data can cause a large change in the model's predictions. High variance usually leads to **overfitting**.

Easy example:

Suppose a model is trained on Training Data A and then on slightly different Training Data B. If the model behaves very differently after this small change, it has high variance.

> **Interview answer:** "Variance tells us how sensitive a model is to the training data. If a small change in training data causes a big change in predictions, the model has high variance. High variance usually leads to overfitting."

2. Overfitting

Simple definition: Overfitting means the model learns the training data too well, including noise and unnecessary details, and performs poorly on new/unseen data.

Easy example: Imagine preparing for an exam by memorizing the exact questions and answers from a practice paper. You score 99% on the practice paper but only 60% on new questions. You memorized instead of understanding. This is similar to overfitting.

Typical pattern:

```
Training Accuracy → Very High
Testing Accuracy → Low
```

> **Interview answer:** "Overfitting happens when a model learns the training data too closely, including noise and unnecessary details. It performs very well on training data but poorly on unseen testing data."

3. Underfitting

Simple definition: Underfitting means the model is too simple and has not learned enough from the training data.

Easy example: Imagine studying almost nothing for an exam. You score 60% on the practice paper and 55% on the new exam. You did not learn enough. This is similar to underfitting.

Typical pattern:

```
Training Accuracy → Low
Testing Accuracy → Low
```

> **Interview answer:** "Underfitting happens when a model is too simple to learn the important patterns in the data. It performs poorly on both training and testing data."

4. Bias

Simple definition: Bias is the error caused by a model being too simple and not learning the important patterns from the data.

Easy example: Suppose you want to predict house prices, but your model only considers house size and ignores location, number of rooms, age, and other important factors. The model is too simple, which causes high bias.

```
High Bias → Underfitting
```

> **Interview answer:** "Bias is the error that happens when a machine learning model is too simple and makes too many assumptions about the data. High bias usually leads to underfitting."

5. Bias vs Variance

Concept	Simple meaning	Related problem
Bias	Model is too simple	Underfitting
Variance	Model is too sensitive to training data	Overfitting
Underfitting	Model learns too little	High Bias
Overfitting	Model learns too much / memorizes	High Variance

6. The Most Important Relationship

```
High Bias → Underfitting
High Variance → Overfitting
```

Think of it as: **Too Simple** → **High Bias** → **Underfitting**
Too Complex / Too Sensitive → **High Variance** → **Overfitting**

7. One-Line Interview Answers

Bias: Bias is the error caused by a model being too simple.

Variance: Variance tells us how sensitive a model is to changes in the training data.

Underfitting: Underfitting means the model has not learned enough from the data.

Overfitting: Overfitting means the model has learned the training data too closely and performs poorly on new data.

8. Super-Easy Memory Trick

```
Bias = Too Simple
Variance = Too Sensitive
Underfitting = Learns Too Little
Overfitting = Memorizes Too Much
```

Note: These notes are simplified for interview preparation. In more advanced ML discussions, bias and variance can be described formally as components of prediction error.